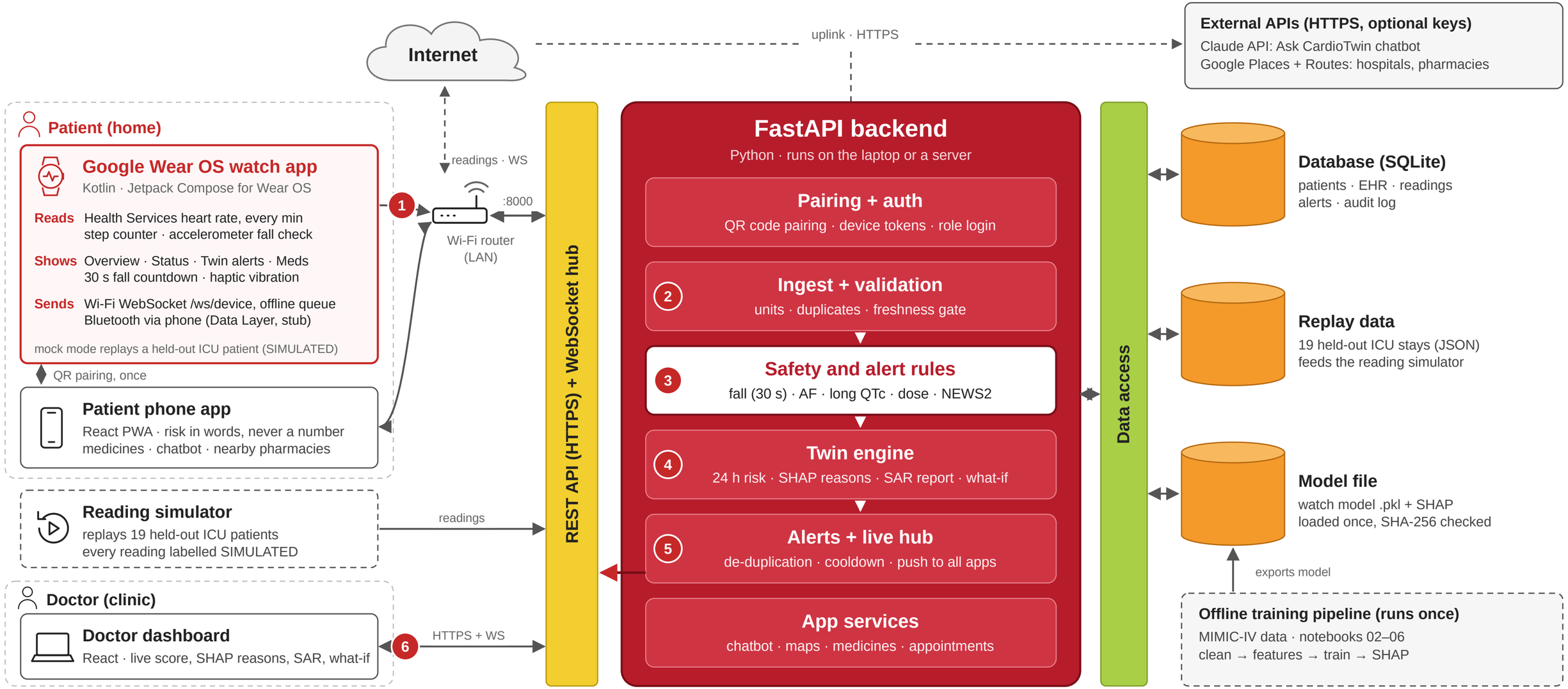
CardioTwin – System Architecture

smartwatch-based cardiac digital twin: from a reading on the wrist to a safe action



Planned next

- Android app · Kotlin, Jetpack Compose
- Samsung Health Sensor SDK: SpO2, ECG
- Phone Bluetooth relay
- PostgreSQL
- Deploy on Render / Railway

Legend

- REST / WebSocket link
- pairing, uplink or offline
- live push to the apps
- 1 to 6 order of the data flow

Ports: backend 8000 · dashboard 5173
phone app 5174 · watch screen 5175

Research prototype trained on MIMIC-IV demo ICU data. Not a medical device.